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(54) **WIRELESS ENHANCED POWER TRANSFER**

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H01F 38/14 (2006.01)

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CPC **H02J 50/12** (2016.02); **H01F 38/14** (2013.01)

(58) **Field of Classification Search**

CPC H02J 50/12; H01F 38/14
See application file for complete search history.

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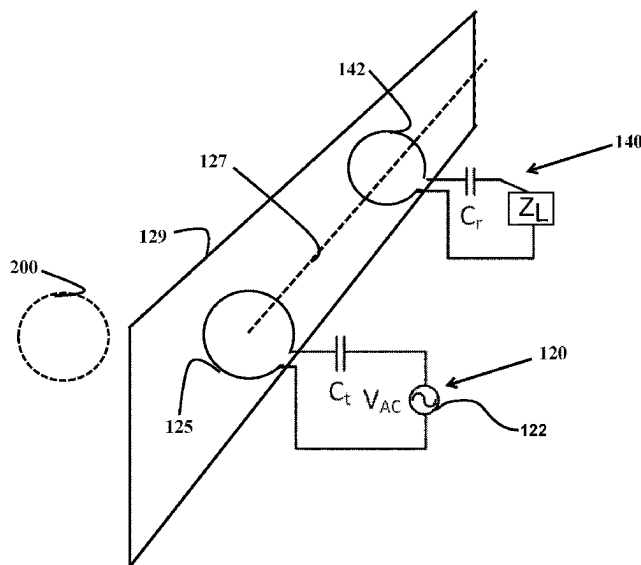
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(57) **ABSTRACT**

A wireless power transfer (WPT) transmitter comprising: a power transmission coil having an axis; a power source operable to generate a time varying current in the power transmission coil that generates a time varying magnetic B field; and an eddy current conductor shaped and positioned to generate eddy currents responsive to the time varying current in the power transmission coil that enhance the B field generated by the time varying current.

12 Claims, 4 Drawing Sheets



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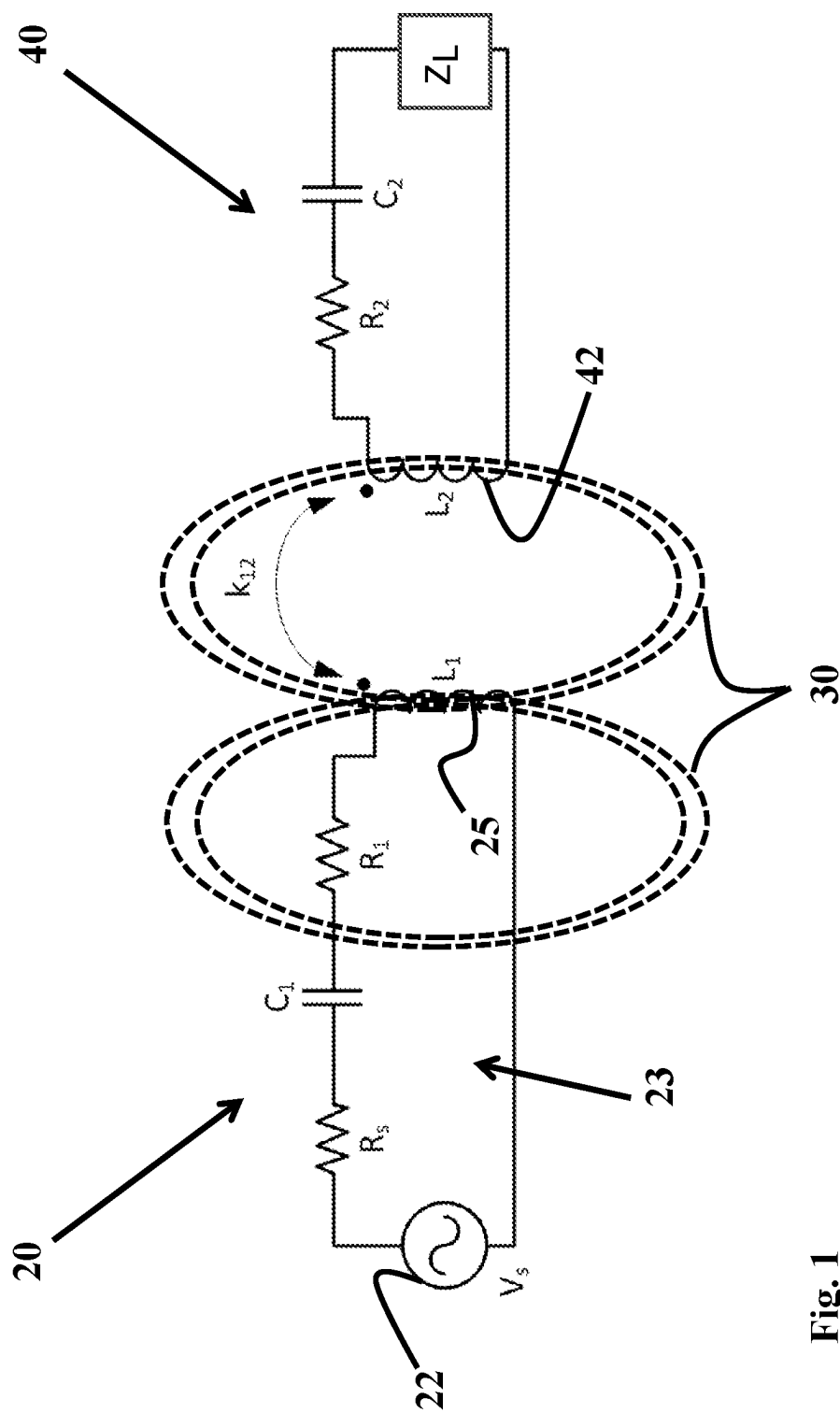


Fig. 1

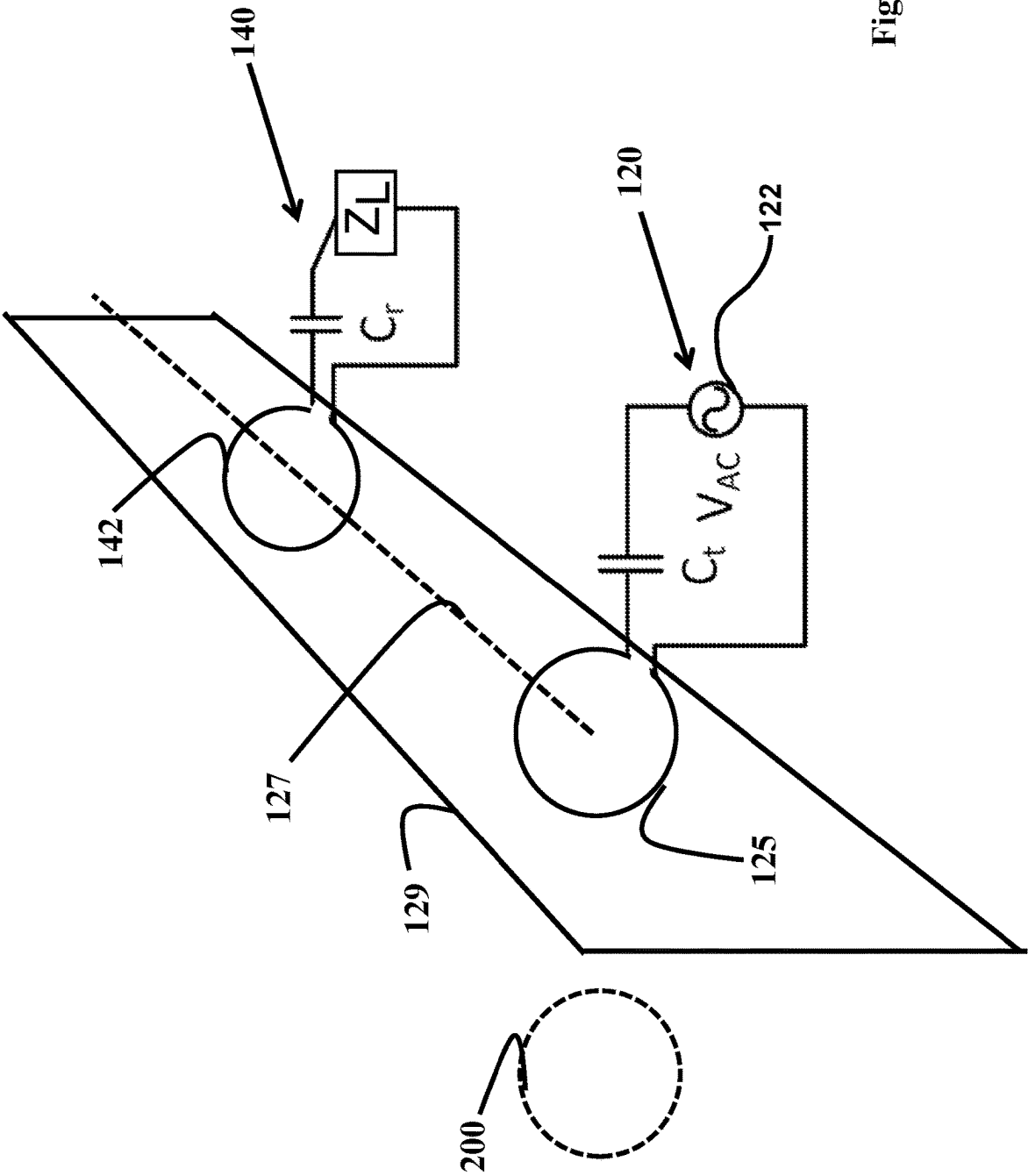


Fig. 2

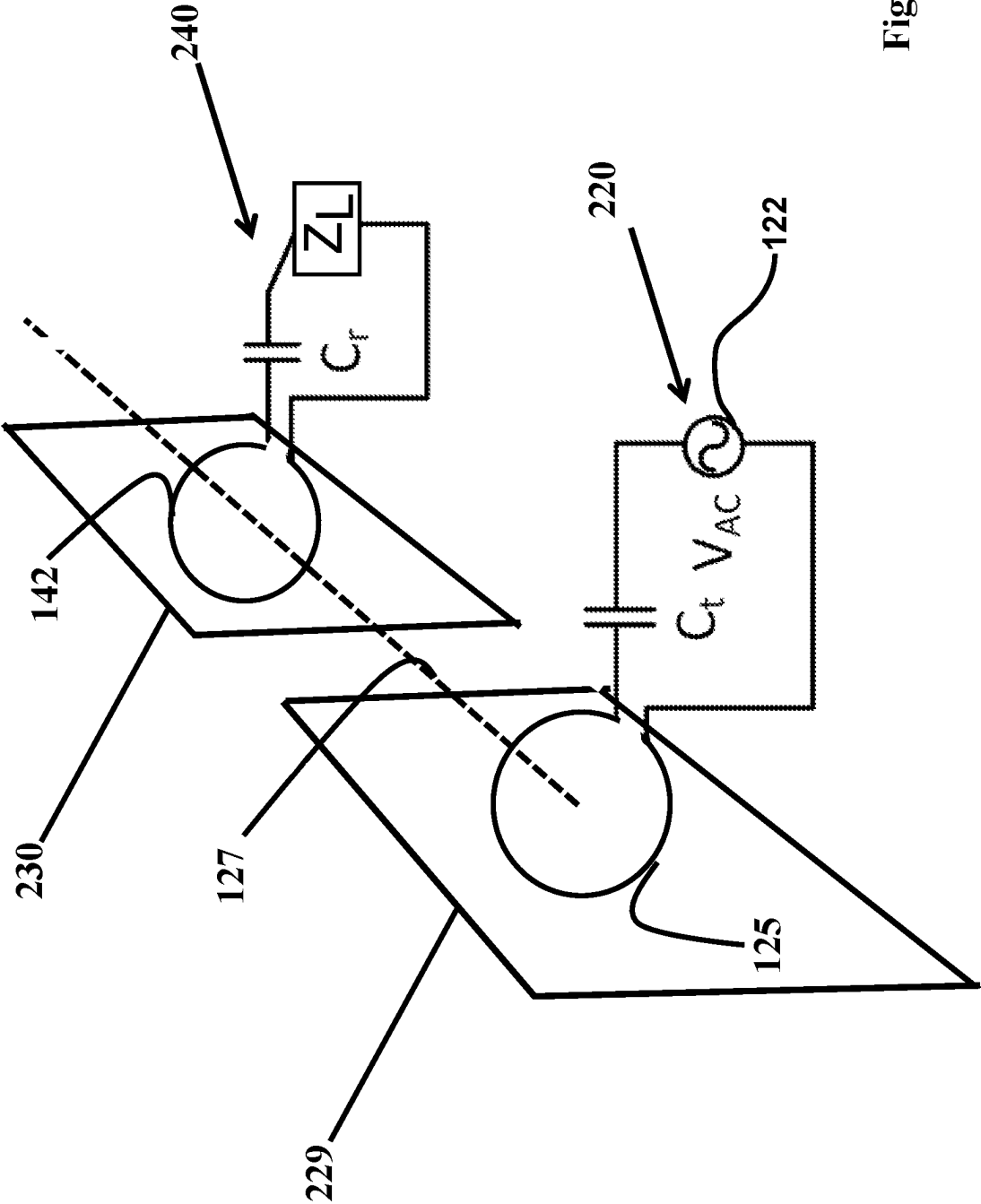


Fig. 3

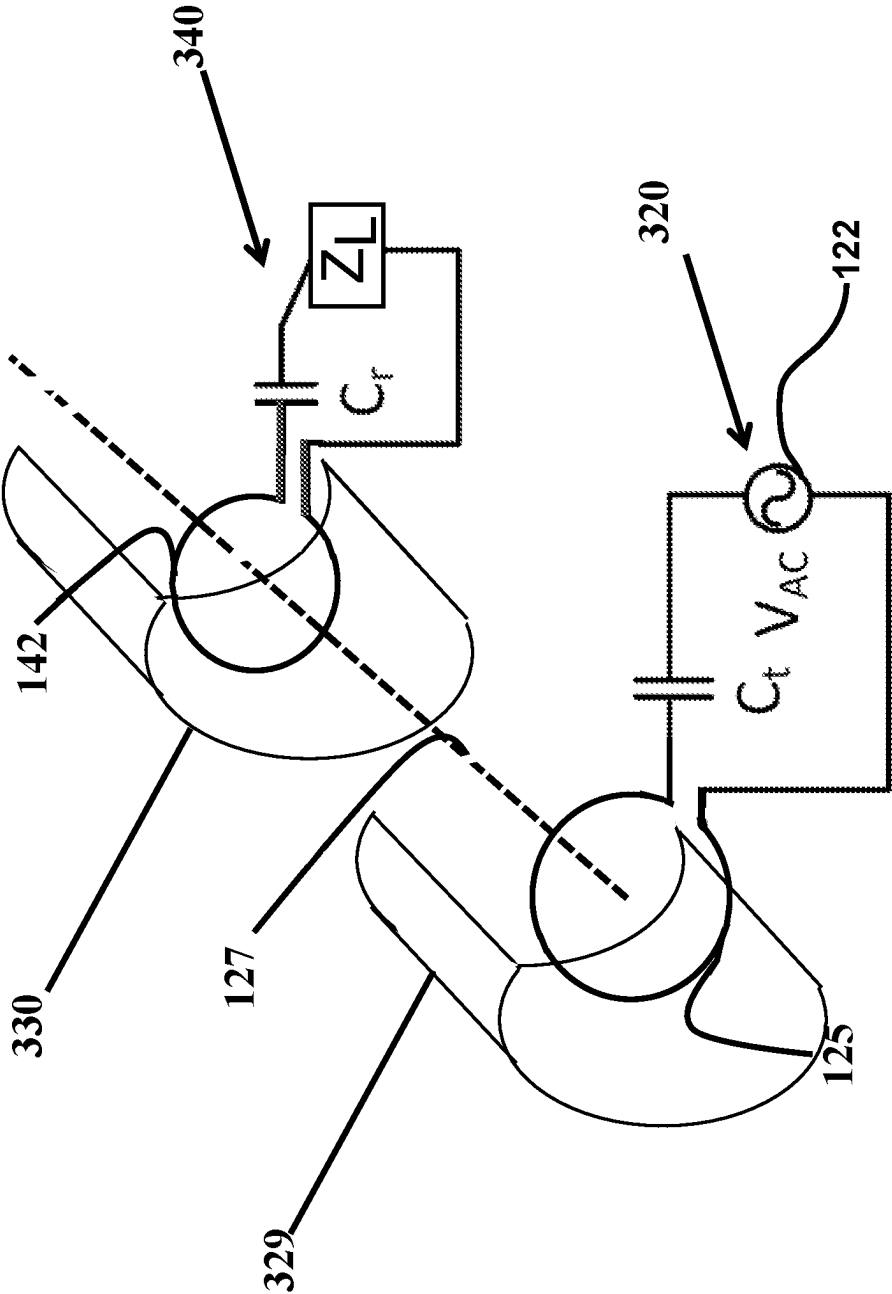


Fig. 4

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WIRELESS ENHANCED POWER TRANSFER

RELATED APPLICATIONS

The present application is a US National Phase of PCT Application No. PCT/IL2020/051055, filed on Sep. 25, 2020 which claims the benefit under 35 U.S.C. 119 (e) of U.S. Provisional Application 62/906,128 filed on Sep. 26, 2019, the disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

Embodiments of the invention relate to wireless power transmission (WPT)

BACKGROUND

Wireless power transmission (WPT) by near field electromagnetic induction is used to wirelessly deliver electric energy to a wide variety of devices that includes by way of example, chargeable batteries in mobile phones, electric tooth brushes, biomedical implants such as pacemakers, electric vehicles, and such devices as Radio-Frequency IDs (RFIDs) and railway balises. Typically, a near field WPT transmitter comprises a resonant energy transmission circuit having a transmission coil and an AC power source. The AC power source drives an AC current at the resonant frequency of the transmission circuit through the transmission coil to generate a time varying magnetic field to which an inductive receiving circuit having a receiving coil may be inductively coupled to extract and store, and/or immediately use, energy from the magnetic field. The strength of the magnetic field generated by the transmission circuit along an axis of the transmission coil decreases with substantially the third power of distance. As a result, energy transfer between the WPT transmitter and the inductive receiving circuit decreases rapidly with distance between the WPT transmitter and the receiving circuit. Resonant inductive WPT systems are therefore generally practical for wireless transmission of energy only over relatively short distances.

SUMMARY

An aspect of an embodiment of the disclosure relates to providing a near field wireless enhanced power transfer system transmitter, which may be referred to by the acronym WEPT, or by WEPT transmitter, that provides enhanced near field transmission of power to electrically powered devices.

In an embodiment, a WEPT transmitter comprises a resonant energy transmission circuit comprising a transmission coil or antenna, generically referred to as a coil, an AC power source configured to drive a time varying current in the transmission coil, and an eddy current, mirror conductor, which may be referred to simply as a mirror conductor. The eddy current mirror conductor is shaped and positioned relative to the WEPT transmission coil to focus a magnetic field generated by current in the transmission coil and form a spatial region, hereinafter also referred to as a magnetic enhancement region, of relatively enhanced field intensity in the magnetic field. For a same consumption of energy, WEPT transmitter in accordance with an embodiment of the disclosure provides improved power transfer to a receiving circuit inductively coupled to the transmitter at the magnetic enhancement region than does a conventional WPT transmitter. In an embodiment the eddy current mirror conductor

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comprises a planar conductor positioned parallel to an axis of WEPT's transmission coil.

In the discussion unless otherwise stated, adjectives such as "substantially" and "about" modifying a condition or relationship characteristic of a feature or features of an embodiment of the invention, are understood to mean that the condition or characteristic is defined to within tolerances that are acceptable for operation of the embodiment for an application for which it is intended.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

BRIEF DESCRIPTION OF FIGURES

Non-limiting examples of embodiments of the invention are described below with reference to figures attached hereto that are listed following this paragraph. Identical structures, elements or parts that appear in more than one figure are generally labeled with a same label in all the figures in which they appear. Labels labeling an icon representing a component or feature in a figure may be used to refer to and identify the component or feature represented by the icon. Dimensions of components and features shown in the figures are chosen for convenience and clarity of presentation and are not necessarily shown to scale.

FIG. 1 schematically shows a conventional resonant inductive WPT transmitter in accordance with prior art transmitting power to a receiving circuit inductively coupled to the WPT transmitter;

FIG. 2 schematically shows a WEPT transmitter comprising a transmission circuit and eddy current mirror conductor, in accordance with an embodiment of the invention;

FIG. 3 schematically shows a WEPT transmitter having an eddy current mirror conductor transmitting power to a receiving circuit comprising an eddy current mirror conductor separate from the WEPT transmitter eddy current mirror conductor, in accordance with an embodiment of the disclosure; and

FIG. 4 schematically shows a WEPT transmitter having a curved eddy current mirror conductor inductively coupled to a receiving circuit also comprising a curved eddy current mirror conductor, in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION

FIG. 1 schematically shows a conventional WPT transmitter **20** being used to transmit power to an inductive power receiving circuit (IPRC) **40**, in accordance with prior art. WPT transmitter **20** comprises an AC power source **22** having an internal resistance R_s and is operable to generate an optionally harmonic time varying voltage V_s . Power source **22** is coupled to a resonator **23** optionally comprising a capacitance C_1 , a resistance R_1 and a power transmission coil **25** having inductance L_1 . Power source **22** and resonator circuit **23** are optionally tuned to operate as a resonant circuit characterized by a resonant frequency f_s and harmonic voltage V_s that varies in time with frequency f_s . Voltage V_s produces a time varying harmonic current I_s in power transmission coil **25** that generates a time varying magnetic field **B** schematically indicated by field lines **30**. "B" is shown in bold script to indicate a vector field

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IPRC **40** comprises a power receiving coil **42** having inductance L_2 with a resistance R_2 optionally coupled in series with capacitance C_2 and a load Z_L which may by way of example be a battery or RFID circuit. Optionally, IPRC **40** operates as a resonant circuit characterized by a same frequency f_s that characterizes WPT transmitter **20**. Receiving coil **42** of IPRC is inductively coupled to transmission coil of WPT transmitter **20** for receiving energy from the WPT transmitter by a magnetic coupling factor k_{12} . Coupling coefficient k_{12} is proportional to an amount of the magnetic field B generated by current I_s in power transmission coil **25** that is encircled by power receiving coil **42**. The power transfer efficiency from WPT transmitter **20** to IPRC **40** is approximately proportional to $(k_{12})^2$.

Along an axis (not shown) of power transmission coil **25**, B generated by current I_s in the loop decreases approximately with the third power of distance from the loop along the axis. As a result, power coupled to IPRC **40** by WPT transmitter **20** generally decreases steeply with distance of IPRC **40** from WPT transmitter **20**.

FIG. 2 schematically shows a WEPT transmitter **120** operating to energize an IPRC **140** in accordance with an embodiment of the disclosure. IPRC **140** is optionally similar to IPRC **40** shown in FIG. 1 and functions substantially as a resonant circuit optionally comprising a power receiving coil **142** in series with a capacitance C_R and a load Z_L , which may by way of example be a battery or RFID circuit.

In accordance with an embodiment, WEPT transmitter **120** functions as a resonant circuit having a resonant frequency f_T and optionally comprises a power transmitting coil **125** having inductance L_T and an axis **127**. Coil **125** is coupled to a capacitance C_T and a power source **122** operable to provide a time varying voltage V_T characterized by the resonant frequency f_T . Voltage V_T drives a current I_T through power transmission coil **125**, which generates a magnetic field B that varies in time with resonant frequency f_T .

In accordance with an embodiment, WEPT transmitter **120** also comprises an optionally planar eddy current, mirror conductor **129** positioned parallel to axis **127**. Current I_T flowing in power transmission coil **125** produces eddy currents in mirror conductor **129** that operate to increase the magnetic field B_T along axis **127** relative to magnitude of a magnetic field that would be present along axis **127** in the absence of eddy current mirror conductor **129**. In an embodiment a magnitude of B_T along axis **127** is greater than that of the magnetic field in the absence of mirror conductor **129** by a field amplification factor equal to or greater than about 1.25. Optionally the field amplification factor is equal to or greater than about 1.5. In an embodiment the field amplification factor is equal to or greater than about 1.75.

Without being limited to any given theory as to how WEPT transmitter **120** creates the amplification factor, it might be considered that eddy currents generated in eddy current mirror conductor **129** responsive to current I_T flowing in power transmission coil **125** produce a magnetic field equivalent to that of a virtual mirror image of I_T on a side of mirror conductor **129** opposite to that of WEPT transmitter coil **125** to produce the field amplification factor. In FIG. 2 the virtual mirror image of I_T , also referred to as an image current of I_T , or simply the image current, is schematically represented by a dashed current loop **200**. The mirror current is an imaginary current that enables determining the magnetic field produced by the eddy currents and current I_T on the side, also referred to as the WEPT side, of mirror conductor **129** where WEPT transmitter coil **125** is located. The image current has a magnitude substantially the same as

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that of I_T (the magnitude of the image current approaches equality with that of I_T as conductivity of conductor **129** approaches that of a perfect conductor) and a direction of flow the same as that of current of I_T —when I_T is clockwise the image current is also clockwise and when I_T is counterclockwise the image current is also counterclockwise. The magnetic field produced by the eddy currents and current I_T on the WEPT side, is the same as that which would be produced by superposition of the magnetic fields generated by I_T and the image current were the image current to be real and mirror conductor **129** absent.

By way of a numerical example, an experimental WEPT transmitter in accordance with an embodiment of the disclosure similar to WEPT transmitter **120** was found to provide an amplification factor of the B field equal to about 1.94. The experimental WEPT transmitter comprised a small multiturn coil and a planar eddy current mirror conductor positioned parallel to and at a distance of about 14 cm (centimeters) from an axis of the coil. The mirror conductor was a half meter by half meter square aluminum plate having thickness equal to about 5 mm (millimeters). When driven by a given current I_T at a frequency of about 1290 Hz, a magnitude of the magnetic B field along the axis at a distance of about 37.5 cm (centimeters) from the power transmission coil was measured to be equal to about 350 pT (picoTeslas). Without the eddy current mirror conductor, the magnetic B field at the same location was measured to be equal to about 180 pT.

Assume, as noted above, that a coupling coefficient k_{12} between the experimental WEPT transmitter to a suitable IPRC is proportional to amplification of the magnetic B field at the IPRC, generated by the WEPT transmitter and that power transfer to the WEPT IPRC is approximately proportional to the square of the coupling coefficient. Then at about 37.5 cm from the experimental WEPT transmitter coil **125** is expected to be about 4 times faster in transmitting power to IPRC **140** than a conventional WPT transmitter similar to WPT transmitter **20**.

Whereas FIG. 2 schematically shows WEPT **120** inductively coupled to IPRC **140** using a single mirror conductor **129** that is adjacent to both the WEPT and the IPRC, embodiments of the disclosure are not limited to coupling a WEPT transmitter to an IPRC using a single mirror conductor that is adjacent to both the WEPT transmitter and the IPRC. By way of example, FIG. 3 schematically shows WEPT **220** inductively coupled to an IPRC **240**, in accordance with an embodiment of the disclosure. WEPT **220** is similar to WEPT **120** but comprises an optionally planar mirror image conductor **229** that does not lie adjacent to IPRC **240** to which WEPT **220** is inductively coupled. In an embodiment IPRC may comprise, as shown in FIG. 3, its own optionally planar mirror image conductor **230** separate from mirror image conductor **229**. Nor is practice of embodiments of the disclosure limited to planar mirror image conductors. By way of example, FIG. 4 schematically shows a WEPT **320** comprising a curved, optionally cylindrical mirror image conductor **329**, in accordance with an embodiment of the disclosure. WEPT **320** is optionally inductively coupled and transmitting power to an IPRC **340** comprising its own curved, optionally cylindrical, mirror image conductor **330**, in accordance with an embodiment.

In the description and claims of the present application, each of the verbs, “comprise” “include” and “have”, and conjugates thereof, are used to indicate that the object or objects of the verb are not necessarily a complete listing of components, elements or parts of the subject or subjects of the verb.

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Descriptions of embodiments of the invention in the present application are provided by way of example and are not intended to limit the scope of the invention. The described embodiments comprise different features, not all of which are required in all embodiments of the invention. Some embodiments utilize only some of the features or possible combinations of the features. Variations of embodiments of the invention that are described, and embodiments of the invention comprising different combinations of features noted in the described embodiments, will occur to persons of the art. The scope of the invention is limited only by the claims.

The invention claimed is:

1. An enhanced wireless power transfer (WEPT) transmitter comprising:

a power transmission coil having an axis;
a power source operable to generate a time varying current I_T in the power transmission coil that generates a time varying magnetic B field; and
at least one eddy current conductor shaped and positioned to generate eddy currents responsive to the time varying current in the power transmission coil that produce an equivalent of a virtual mirror image of I_T which enhances the time varying magnetic B field generated by the time varying current;
wherein the at least one eddy current conductor extends alongside the axis and is absent slits or holes that may operate to distort the virtual mirror image of I_T .

2. The WEPT according to claim 1 wherein the at least one eddy current conductor comprises a planar eddy current

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3. The WEPT according to claim 2 wherein a plane of the planar eddy current conductor is substantially parallel to the axis of the power transmission coil.

4. The WEPT according to claim 1 wherein the time varying magnetic B field is enhanced by a field amplification factor equal to or greater than about 1.25.

5. The WEPT according to claim 4 wherein the time varying magnetic B field is enhanced by a field amplification factor equal to or greater than about 1.5.

6. The WEPT according to claim 5 wherein the field amplification factor is equal to or greater than about 1.75.

7. The WEPT according to claim 6 wherein the field amplification factor is equal to or greater than about 2.

8. The WEPT according to claim 1, wherein the at least one eddy current conductor comprises an eddy current conductor at least a portion of which partially surrounds the axis of the power transmission coil.

9. The WEPT according to claim 8 wherein the at least one eddy current conductor that partially encircles the axis of the power transmission coil is cylindrical.

10. The WEPT according to claim 1, wherein the at least one eddy current conductor comprises a plurality of eddy current conductors.

11. The WEPT according to claim 1, wherein the at least one eddy current conductor is shaped and positioned to lie adjacent to an inductive power receiving circuit (IPRC) to which the WEPT transmits power.

12. A system comprising:

the WEPT according to claim 1; and

an inductive power receiving circuit (IPRC) that is inductively coupled to the WEPT to receive power from the WEPT.

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